Faculty of Science
Department of Mathematics
VU

Linear Algebra and Calculus
Resit Exam 2025
10-02-2026, 08:30–11:15

Name: ______________________________

Student ID: ______________________________ **Grade:** ______

Question:	1	2	3	4	5	6	7	8	9	10	Total
Points:	2	6	2	8	7½	5	3	4	4½	3	45
Score:											

$$\text{Calculus Grade} = \frac{\#\text{points of questions (1 - 4)}}{2} + 1 = \_\_$$

$$\text{Linear Algebra Grade} = \frac{\#\text{points of questions (5 - 10)}}{3} + 1 = \_\_$$

Final Grade = Calculus Grade * 0.3 + Linear Algebra Grade * 0.7

Read the instructions carefully!

- The use of a calculator, book, or notes is **not** allowed.
- Answer the questions in the spaces provided with black or blue pen. If you run out of room for an answer, continue on the back of the page, but make sure the final answer starts on the front.
- You can ask for extra paper to do your calculations, but you shall not submit it, the graded answers will be the ones on these papers, so make sure the final answers are properly explained here, starting in the blank space provided and finishing on the back if necessary.
- Answers without supporting work will not be given full credit, so make sure to explain the theoretical basis of what you do for a full score.
- If you feel like you are short on time, prioritize questions you know or those with higher score count.

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1. **(2 pts.) Calculus:** Compute $\lim_{x \to 0} \frac{\sin x - x}{x^3}$.

   Hint: Close to $x = 0$ the sine function can be written as $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + O(x^6)$, where $O(x^6)$ is a sum of terms of degree 6 or higher.

2. **Calculus:** Define

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$$

(a) **(3 pts.)** Show that f is continuous at $x = 0$.
Hint: You can use the Squeeze Theorem.

(b) **(3 pts.)** Determine whether f is differentiable at $x = 0$.
Hint: Recall that $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$.

3. **(2 pts.) Calculus:** Consider the system of functions $F = (f_1, f_2)$ where $f_1(x, y) = \dfrac{x^2 + y}{y + 1}$ and $f_2(x, y) = x \sin(xy)$. Compute the Jacobian of F at any point (x, y).

4. **Calculus:** Consider the function $f(x) = e^{-x} - x$.

(a) **(3 pts.)** Show that f has one root in the interval $[0, 1]$.

(b) **(1½ pts.)** Compute $\lim_{x \to \pm\infty} f(x)$.

(c) **(1½ pts.)** Compute the derivative of $f(x)$ for all x in $\mathbb{R}$.

(d) **(2 pts.)** Does f have any other root besides the one found in part a)? Justify your answer.

5. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear map defined by

$$T(x, y, z, w) = \begin{bmatrix} x + y + w \\ y + z \\ x + z + w \end{bmatrix}.$$

(a) **(1 pts.)** Give the standard matrix A of T.

(b) **(2½ pts.)** Give a basis for the null space of A.

(c) **(2 pts.)** Give a basis for the column space of A.

(d) **(1 pts.)** Determine the rank and nullity of T.

(e) **(1 pts.)** Is T injective? Is it surjective? Justify your answer.

6. Let A, B be two square matrices of real numbers with A invertible.

   (a) **(1 pts.)** Solve the matrix equation $AXA^{-1} = B$ for X.

   (b) **(2 pts.)** Show that if B is diagonalizable, then X is diagonalizable.

   (c) **(2 pts.)** Show that X and B have the same eigenvalues.

7. **(3 pts.)** Let $C^1([0,1])$ be the set of differentiable functions on the interval $[0,1]$ and let $V = \{f \in C^1([0,1]) \mid f(0) = f(1) = 0\}$ be the subset of $C^1([0,1])$ of functions such that satisfy the conditions $f(0) = 0$ and $f(1) = 0$. Show that V is a vector subspace of $C^1([0,1])$.

8. Let $A = \begin{bmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \alpha \in \mathbb{R}$.

(a) **(1 pts.)** Find all eigenvalues of A.

(b) **(3 pts.)** For which values of α is A diagonalizable? Justify your answer.

9. Consider the following vectors in $\mathbb{R}^3$;

$$v_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

(a) **(3 pts.)** Find an orthogonal basis for $\text{Span}\{v_1, v_2, v_3\}$.

(b) **(1½ pts.)** Normalize the basis from a) to obtain an orthonormal basis.

10. Mark each of the following statements as true or false. If the statement is true, give a proof. If the statement is false, give a proof or provide a counterexample.

(a) **(1 pts.)** If $\underset{C \leftarrow B}{P}$ is the change of basis matrix from a basis B to a basis C, then $\det \underset{C \leftarrow B}{P} = 0$.

(b) **(1 pts.)** If $\mathbb{P}_n$ is the space of polynomials of degree $\leq n$, then $\dim \mathbb{P}_n = n$.

(c) **(1 pts.)** If u, v are vectors in $\mathbb{R}^n$ and $u \cdot v = 0$, then either u or v is the zero vector.